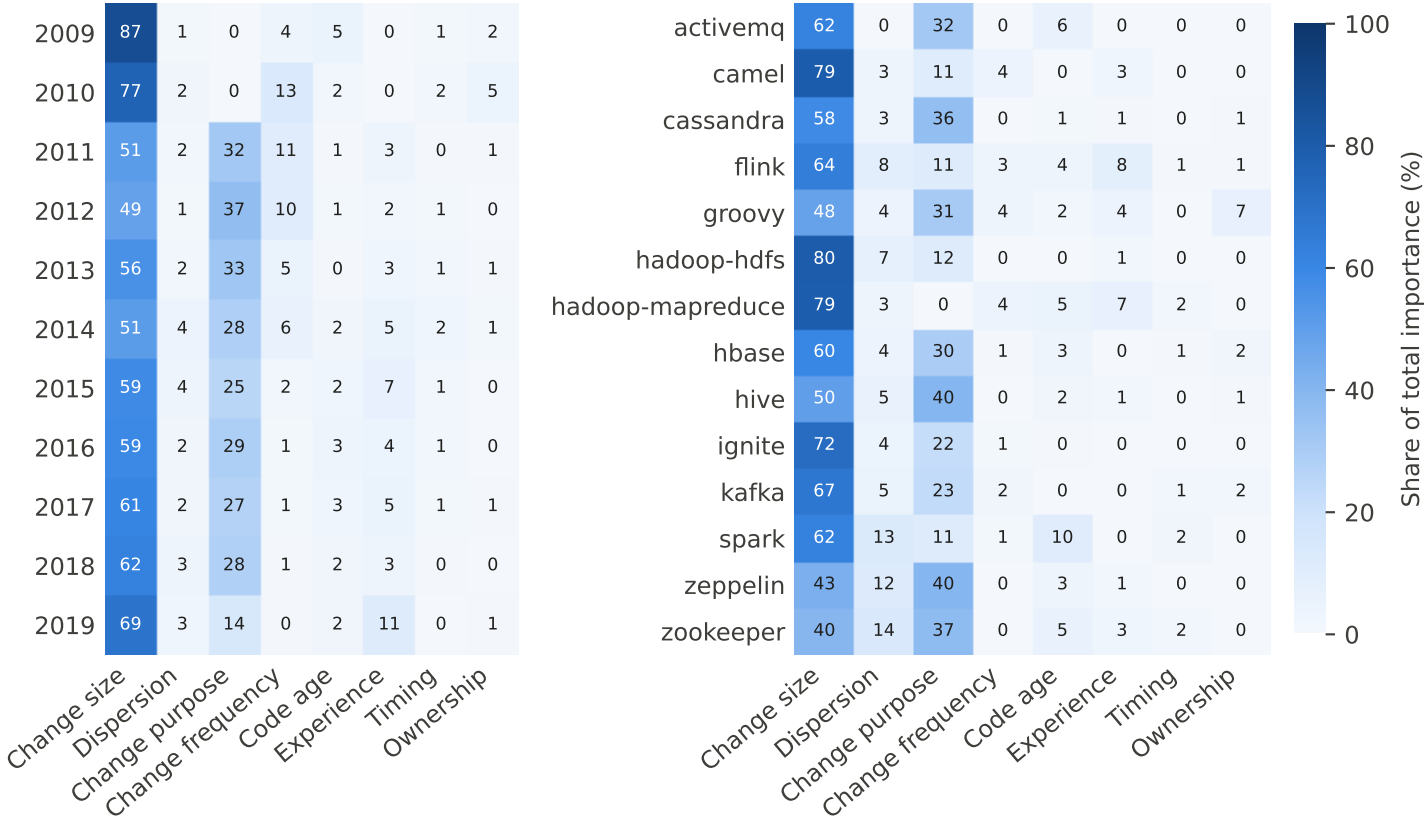
(a) Test year (train: preceding 3 years)
2009 87 1 0 4 5 0 1 2
2010 77 2 0 13 2 0 2 5
2011 51 2 32 11 1 3 0 1
2012 49 1 37 10 1 2 1 0
2013 56 2 33 5 0 3 1 1
2014 51 4 28 6 2 5 2 1
2015 59 4 25 2 2 7 1 0
2016 59 2 29 1 3 4 1 0
2017 61 2 27 1 3 5 1 1
2018 62 3 28 1 2 3 0 0
2019 69 3 14 0 2 11 0 1
Change size
Dispersion
Change purpose
Change frequency
Code age
Experience
Timing
Ownership
(b) Project (chronological 70/30 split)
activemq 62 0 32 0 6 0 0 0
camel 79 3 11 4 0 3 0 0
cassandra 58 3 36 0 1 1 0 1
flink 64 8 11 3 4 8 1 1
groovy 48 4 31 4 2 4 0 7
hadoop-hdfs 80 7 12 0 0 1 0 0
hadoop-mapreduce 79 3 0 4 5 7 2 0
hbase 60 4 30 1 3 0 1 2
hive 50 5 40 0 2 1 0 1
ignite 72 4 22 1 0 0 0 0
kafka 67 5 23 2 0 0 1 2
spark 62 13 11 1 10 0 2 0
zeppelin 43 12 40 0 3 1 0 0
zookeeper 40 14 37 0 5 3 2 0
Change size
Dispersion
Change purpose
Change frequency
Code age
Experience
Timing
Ownership
Share of total importance (%)
100
80
60
40
20
0